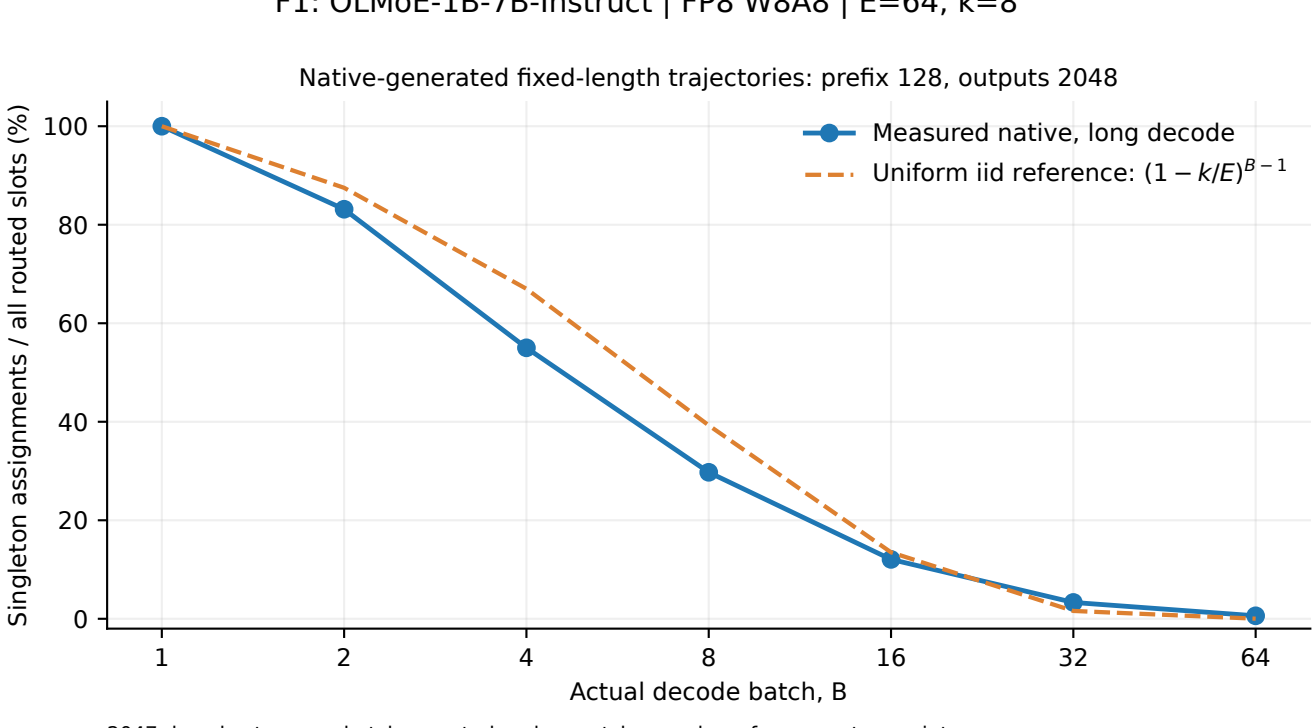
F1: OLMoE-1B-7B-Instruct | FP8 W8A8 | E=64, k=8
Native-generated fixed-length trajectories: prefix 128, outputs 2048
Measured native, long decode
Uniform iid reference: $(1 - k/E)^{B-1}$
Singleton assignments / all routed slots (%)
100
80
60
40
20
0
1
2
4
8
16
32
64
Actual decode batch, B
2047 decode steps per batch; counted replay matches each performance transcript.
1 source execution(s) per batch; tokens/layers are not independent runs. No skew fit or confidence interval.
EOS ignored; occupancy is neither natural task quality nor a physical DRAM/time saving.